

OPERATING SYSTEMS

Process Management and Scheduling

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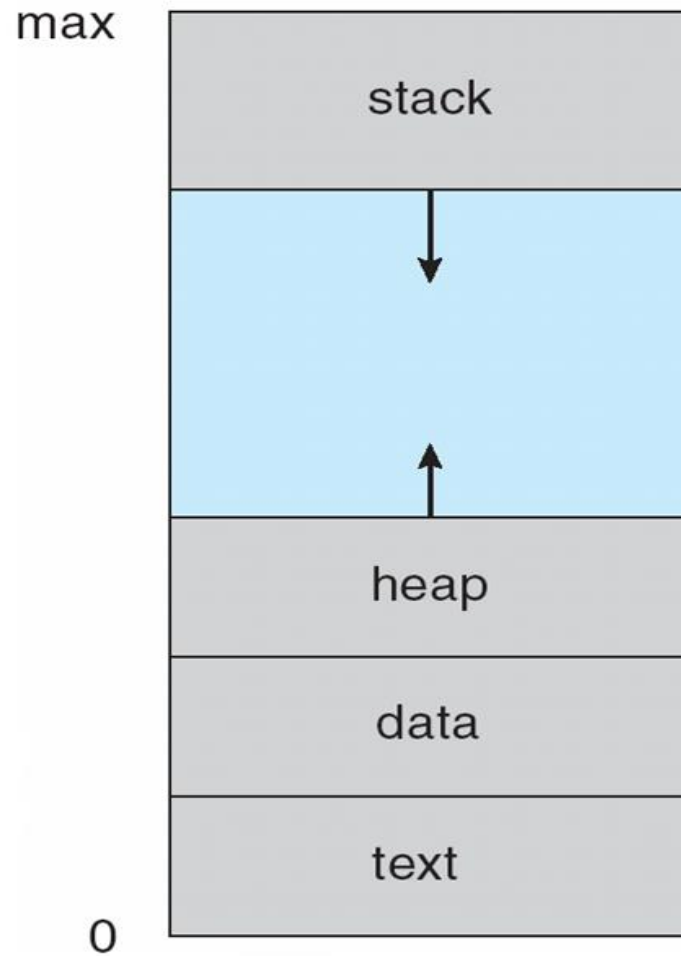
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- Process - An instance of **computer program in execution**
- Some times also called as jobs or task. In
 - Batch system – **jobs**
 - Time-shared systems – **user programs or tasks**
- Process can have several threads of execution or several different pieces of the program running in parallel
- Each process has its own private resources, data and associated statistics
- A OS creates a data structure in the memory

- In memory, a process consists of **multiple parts**:
 - **Program code**, also called **text section**
 - **Stack** containing temporary data such as Function parameters, return addresses, local variables
 - **Data section** containing global variables
 - **Heap** containing memory dynamically allocated during run time

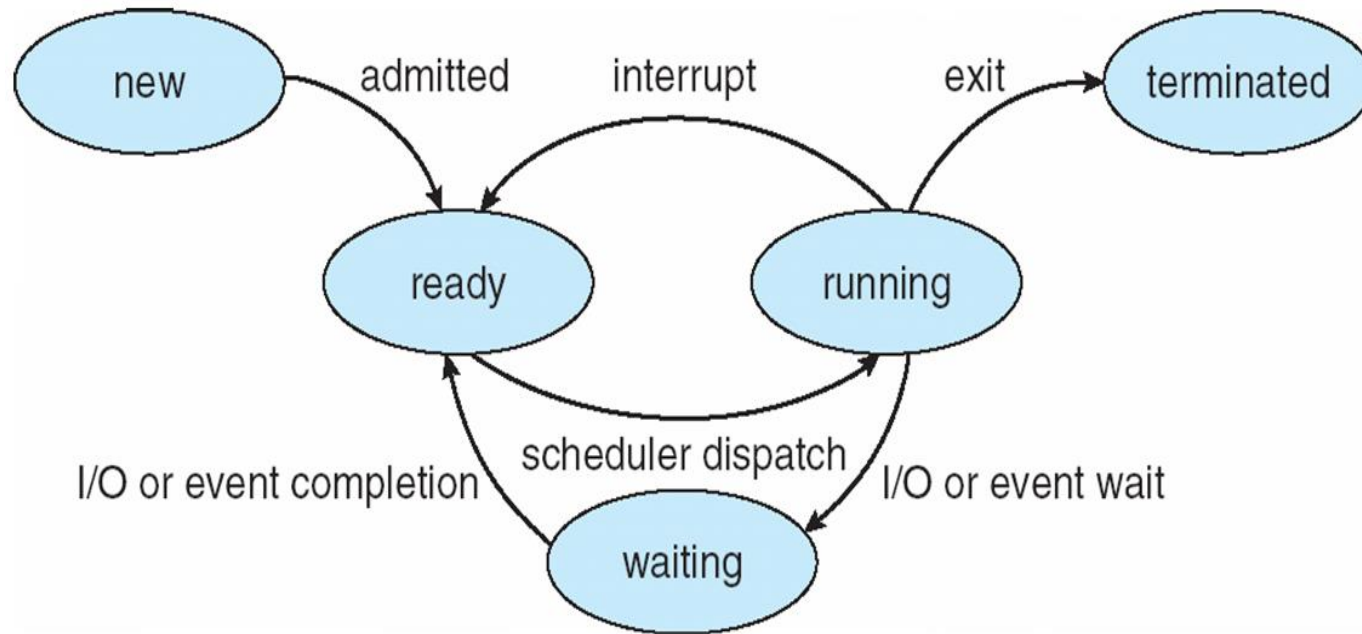
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Process Concept



As a process executes, it changes state

- **new:** The process is being created
- **ready:** The process is waiting to be assigned to a processor
- **running:** Instructions are being executed
- **waiting:** The process is waiting for some event to occur
- **terminated:** The process has finished execution

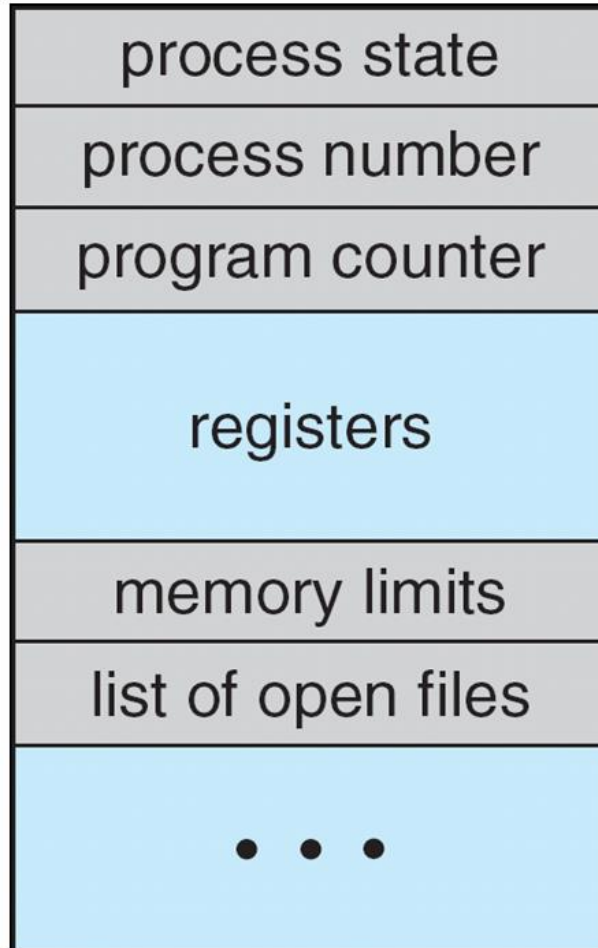


Information associated with each process

- **Process state** – new, ready, running...
- **Process Number** – process id, parent process id
- **Program counter** – next instr. for exec
- **CPU registers** – no. and type of reg.used
- **Memory-management information**
 - Limit of page table and segment table
- **Accounting information**
 - Amt of CPU time used, time limit
- **I/O status information**
 - List of I/O devices

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Process Control Block





THANK YOU

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